Status	Finished
Started	Sunday, 18 October 2026, 9:12 PM
Completed	Sunday, 18 October 2026, 9:18 PM
Duration	5 mins 57 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

a. It builds a sorted array one element at a time
b. It divides the list into two halves, sorts each half, and then merges them
c. It repeatedly finds the minimum element and moves it to the sorted part of the list
d. It compares adjacent elements and swaps them if necessary

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **2**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

a. Quick Sort
b. Merge Sort
c. Heap Sort
d. Bubble Sort

The correct answer is: Bubble Sort

Question **3**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

a. sort()
b. arrange()
c. order()
d. sorted()

The correct answer is: sort()

Question **4**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

- a. Arranging data in a particular format
- b. Searching for data in a list
- c. Inserting data into a list
- d. Deleting data from a list

The correct answer is: Arranging data in a particular format

Question **5**

Complete

Mark 0.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

- a. Bubble Sort
- b. Linear Search
- c. Quick Sort
- d. Merge Sort

The correct answer is: Merge Sort

Question **6**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

- a. It is based on the divide-and-conquer approach
- b. It sorts data using a single pass
- c. It does not require recursion
- d. It is less efficient than bubble sort

The correct answer is: It is based on the divide-and-conquer approach

Question **7**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 40,70,80,90,120,150,200,300,400,470
- b. 200,470,80,150,40,90,300,400,70,120
- c. 40,80,90,150,200,300,400,470,70,120
- d. 80,150,200,470,40,90,300,400,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **8**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

- a. By comparing elements sequentially
- b. By breaking the input into smaller parts and merging them
- c. By using the bubble-up method
- d. By sorting data in a single pass

The correct answer is: By breaking the input into smaller parts and merging them

Question **9**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

- a. A sorting algorithm is stable if it preserves the order of duplicate keys
- b. A sorting algorithm is stable if it preserves the order of non-duplicate keys
- c. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys
- d. A sorting algorithm is stable if it preserves the order of all keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **10**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Sorting the entire list sequentially
b. Combining sorted sublists
c. Comparing each element with the others
d. Dividing the input into smaller subproblems

The correct answer is: Dividing the input into smaller subproblems

Question **11**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

a. Selection
b. Distribution
c. Insertion
d. Extraction

The correct answer is:
Insertion

Question **12**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

a. It recursively sorts subproblems
b. It uses the heap data structure
c. It divides the list into sublists
d. It bubbles up the largest element in each pass

The correct answer is: It bubbles up the largest element in each pass

Question **13**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

a. Selection
b. Duplicates
c. Searching
d. All of the above

The correct answer is: All of the above

Question **14**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

a. It only works with integer arrays
b. It requires external libraries
c. It sorts data out of the box efficiently
d. It is less efficient than custom sorting algorithms

The correct answer is: It sorts data out of the box efficiently

Question **15**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

a. It allows for quicker identification of duplicates
b. It has no effect on the analysis process
c. It complicates the analysis process
d. It makes the analysis slower

The correct answer is: It allows for quicker identification of duplicates